A Fast Load-shedding Algorithm for Power System based on Artificial Neural Network

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***Abstract*—When the power system breaks down or the unit is shut down due to equipment maintenance, emergency control measures such as load cutting should be taken to ensure the stable operation of the system. With the expansion of power grid scale and the complexity of power grid structure caused by new energy grid-connection, the traditional physical modeling method is difficult to accurately reproduce the characteristics of the power grid, and it is impossible to get an effective load cutting scheme in a short time. From the perspective of data analysis, this paper analyzes and sorts out the operation environment data of power grid, and generates the sample data set for model training. Based on the sample data, the artificial neural network was trained, and the potential operation characteristics of the power grid were extracted by means of supervised learning, so as to quickly obtain the load shedding schemes under different operation states of the power grid. The effectiveness of the proposed method is verified by comparing the proposed method with the traditional one based on an example system.**

Keywords—Load cutting, Emergency control, Data analysis, Artificial neural network

I. INTRODUCTION

When the power system fails or is disturbed, it will not immediately affect the power balance of the system, but will cause some abnormal electrical quantity, such as line overload, bus overvoltage and others which lead the system enter an emergency state. The power system in emergency can still be operated for a short time, but emergency control measures should be taken to avoid further expansion of the influence of faults and disturbances. Emergency control of power grid includes two stages: selective fault removal and prevention of accident expansion[1]. In the first stage, cutting machine, cutting load and other measures are generally adopted to remove components quickly to eliminate faults and alleviate disturbances. When the system is still in an emergency state after the removal of components, it is necessary to adopt methods such as disassembly to avoid chain failures and system collapse[2]. It can be seen that if the system can be restored to the normal state in the first stage of emergency control, the impact on users can be minimized. The load removal is more flexible and more suitable for practical application.

In [3], the augmented power flow calculation that takes into account the variation of power network frequency before and after load shedding is carried out to coordinate the effectiveness and economy of load shedding control. Reference [4] proposed the "micro-step discrete method", and combined with the physical significance of gradient to transform the load shedding problem into the optimal control problem; Reference [5] considered the sensitivity of the shear load to the transient stability margin, and adopted multi-initial value optimization to improve the convergence of the algorithm. All the above algorithms set up physical models to simulate from the characteristics of power network. With the increasing complexity of the grid structure, it will be more and more difficult to build the model, and it will take a long time to get the load cutting result through the model. The data analysis method gradually developed in recent years can directly analyze the power grid operation data and obtain the potential rules[6] in order to avoid the information error caused by the simplification and approximation of the simulation process, which has become a feasible means to obtain the emergency control strategy.

As a kind of data analysis method, neural network is widely used in power industry due to its strong applicability. At present, neural network is mainly used in load forecasting[7] and power behavior analysis[8] in power system, but it is not widely used in emergency control. Based on the idea of data analysis, this paper selects appropriate data of power grid operation environment to train the artificial neural network, realizes the direct extraction of potential characteristics of the power grid, and obtains the load cutting control strategy more quickly.

II. OPTIMIZATION OF ARTIFICIAL NEURAL NETWORK PERFORMANCE

Artificial neural network (ANN), also known as neural network, is a research hotspot in the field of artificial intelligence in recent years[9]. Its essence is an operational model. Neural network can be divided into feedforward network and feedback network according to the model structure, and can be divided into supervised learning and unsupervised learning. The neural network selected in this paper is a feedforward network under supervised learning. As a mathematical model of algorithm, the performance of neural network is superior in training accuracy and speed. The performance of neural network is mainly affected by the following factors:

A. Loss function

Loss function, also known as cost function, is the objective function of neural network optimization, and the

objective of neural network training process is to minimize the loss function. The smaller loss function means that the predicted result is close to the real one and the network performance is better. Therefore, the selection of appropriate loss function will directly affect the iterative process of the network and the precision of the final generation of the network. Common loss functions include mean squared loss function, cross entropy loss function, etc[10]. The calculation formulas corresponding to the square loss function and cross entropy loss function are shown in equation (1)(2):

$$C_{Sq}=\frac{1}{2n}\sum_{x}\left\|y(x)-a^{L}(x)\right\|^{2} \quad (1)$$

$$C_{Cr}=-\frac{1}{n}\sum_{x}[y\ln a+(1-y)\ln(1-a)] \quad (2)$$

Where, C_{Sq} and C_{Cr} represent the square loss and cross entropy loss respectively, x represents the characteristic quantity, y represents the label quantity, a represents the predicted value, and n represents the number of samples. Since the cross entropy loss function satisfies that the higher the error (α-y) is, the higher the gradient is, so that the parameters can be updated more quickly in the training process and the iteration effect is better. The characteristic of cross entropy loss function makes it focus on the correctness of classification (maximum eigenvalue) and less on the numerical distribution of misclassification (other than maximum eigenvalue). This feature makes it better for classification problems, but not for regression problems.

B. Optimization algorithm

The purpose of the optimization algorithm is to minimize (or maximize) the network's objective function (loss function) by improving the training method. By adjusting the algorithm, the convergence speed of the network can be accelerated, and the influence of local minimum and saddle point on the iterative process can be reduced. The optimization algorithm is divided into first-order optimization algorithm and second-order optimization algorithm. Both gradient descent and stochastic gradient descent are first-order optimization algorithms. Different from the iteration direction of first-order optimization by updating the current position gradient of the independent variable, the second-order optimization algorithm uses the second-order derivative and introduces factors such as momentum to make the independent variable approach to the optimal solution more rapidly. However, the use of the second-order derivative increases the calculation cost, so it is generally only used for the training of deep neural network.

III. SELECTION OF SAMPLE DATA SETS

A. Selection of features

The influence of disturbance can be eliminated by selecting load-shedding node and degree reasonable[11]. Selected features of sample data should reflect the power grid running environment completely.

The output of each generation node of the system, the load rate of each line and the load level of the load node are selected as the sample characteristic quantity. The above electricity quantity reflects the operation status of power network from three aspects of power generation, power transmission and power consumption. The potential connection between it and load cutting strategy can be extracted through the training of sample data, and the mapping can be realized through the obtained network.

Therefore, the number of features in the sample data of neural network is:

$$n_{features}=\sum_{i=1}^{n_G}n_U^i+n_B+n_{\mathrm{L}} \quad (3)$$

Where, n_G is the number of generating points of the system, n_U^i is the number of generating units of the ith generating node, n_B is the number of system branches, and n_L is the number of system load nodes.

B. Selection of labels

Spervised learning analyzes new situations by mining the potential association between existing feature quantities and tag quantities. Load shedding strategy in the actual operation of power grid is selected as the sample label quantity for training, and the training results will be more in line with the actual operation law. However, the grid's historical data is often hard to obtained. It is considered to obtain the system load shedding strategy under random disturbance through simulation, and generate the final sample data set by taking the corresponding grid environment information as the characteristic quantity. In order to facilitate the model training, the sample data are normalized.

The decision-making process of quick load cutoff is shown in the figure 1.

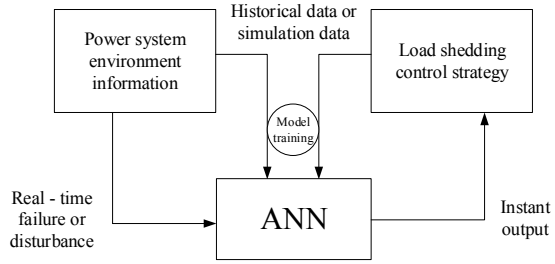


Figure 1. Quick load shedding strategy flow diagram

IV. EXAMPLE ANALYSIS

A. Model structures

This paper takes the IEEE RST-79 system as an example and builds a model based on the MxNet deep learning framework[12] for simulation calculation. Compared with other frameworks, MxNet framework has the advantages of faster development speed, stronger

portability, simplicity and modularity, and is more suitable for the application of machine learning. The environment information and corresponding load shedding strategy are obtained through simulation when random disturbance occurs. The system wiring diagram is shown in figure 2 and the final sample data structure is shown in table 1.

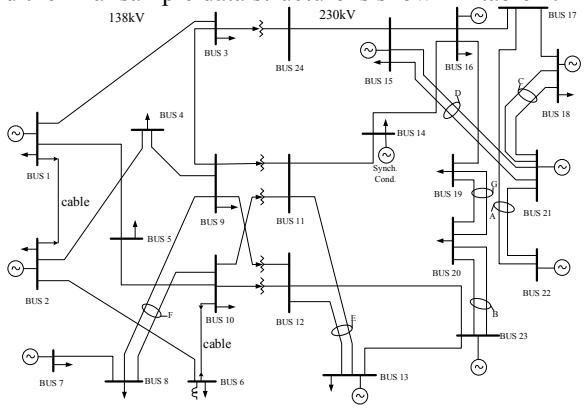


Figure 2. " IEEE RST-79 system wiring diagram

TABLE I. " SAMPLE DATA STRUCTURE TABLE

	Sample size	The number of features	The number of labels
Training sample	80000	87	17
Test sample	20000	87	17

The 4-layer network model with the best effect is finally selected through results comparing. The dropout layer is introduced after the full connection layer except the output layer, and its role is to randomly discard neurons in the hidden layer in order to prevent overfitting. The initial learning rate of the model was set at 0.1, and it decayed to 0.9 times of itself after 50 iterations. The model structure is shown in figure 3.

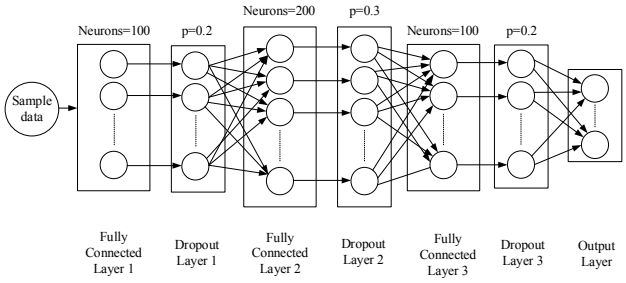


Figure 3. " Example neural network structure diagram

B. *Results analysis*

The training results of the network are shown in figure 4. As can be seen from figure 4, with the increase of the number of iterations, the network's training loss and test loss keep decreasing, and tend to be flat near the 600 iterations. The model basically converges and meets the error requirements. The difference between training loss and test loss is not large, which indicates that the model has not been fitted and has good universality.

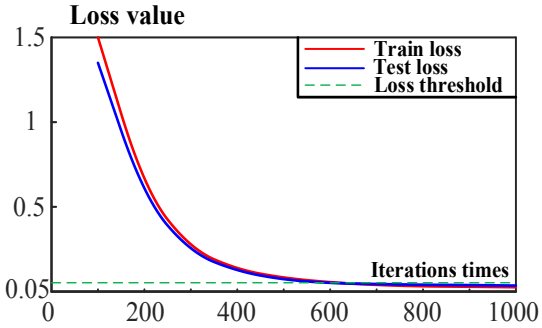


Figure 4. " Annotation of this figure.

1) *Validity test*

Since the model in this paper deals with a regression problem, the accuracy of the model is not as easy to express as the classification problem. If we set the absolute value difference between any output item (17 items in total) of the model and the actual value within 0.01, the model result is considered to be correct, then the accuracy rate of the model in this paper is 98.2%.When the error threshold is set to 0.005, the model accuracy rate becomes 83.6%.This indicates that the accuracy of the load shedding strategy obtained by the model in this paper is at most 1%.This is because the implicit information and model structure of the training data are not enough to support better results. However, for practical problems, the realized accuracy of load shedding is sufficient.

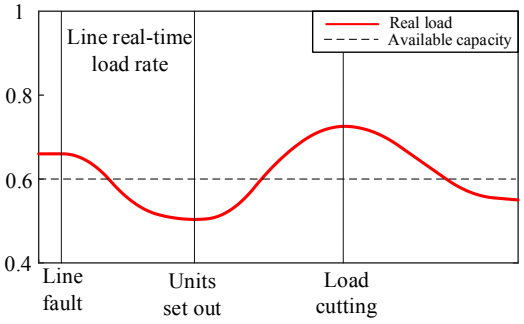


Figure 5. " Line 8-9 power flow curve

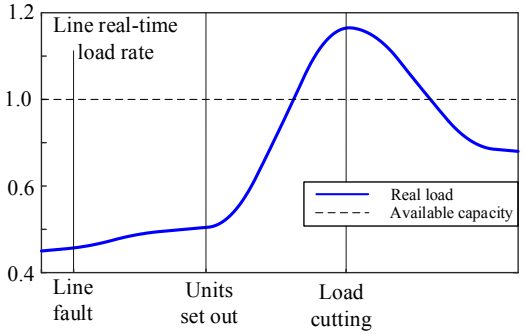


Figure 6. " Line 8-10 power flow curve

A simple disturbance event is selected to verify the effectiveness of the load shedding strategy. Suppose that

due to the line fault, the rated capacity of the branches between 8.9 nodes is reduced to 60% of the original. At this time, the power supply pressure in the area around node 7 and 8 increases, and the system is prone to over-limit. The power network information in this state is input into the trained ANN model, and the corresponding load cutting strategy is 20.0% load cutting of node 7 and 81.4% load cutting of node 8. The above control strategy is adopted and the power flow change curves of branch 8-9 and branch 8-10 are shown in figure 5 and figure 6.

As can be seen from the figure, when the disturbance (unit withdrawal) occurs, the inequality constraint of the power network is broken, and both lines run under excessive load. After the control strategy is executed, the power flow of both branches drops to within the available capacity, and the system returns to normal operation after leaving the emergency state. Therefore, the neural network model in this paper can obtain the accurate load - shedding strategy.

2) Speed test

In order to facilitate the comparison and ensure the authenticity of the verification results, the traditional simulation model of system load shedding was built in the python environment. The timing module is introduced in both the simulation model and the neural network model. The training time under the error requirement of the statistical neural network and the time of the load cutting strategy of the two models are obtained after the given characteristic quantity. In order to ensure the availability of the network, the iteration number of the neural network is set to 800. Specific time-consuming information is shown in table 2.

TABLE II. DIFFERENT MODELS'S CALCULATE TIME TABLE

	Model training time	**Result Outputting time**
Traditional simulation model	—	0.183 sec
ANN	6 min	0.021 sec

As can be seen from table 2, although the neural network needs a certain amount of time to train the network, the time to obtain load cutting strategy through the neural network is greatly reduced compared with the traditional simulation model. The solution process of the traditional simulation model needs to set many constraints according to the actual situation of the power grid, and each solution needs to repeat the iterative calculation process. The more complex the power grid is, the longer its solution time will be. The training process of the model is the main time-consuming step for ANN. Once the model is trained successfully, the corresponding control strategy can be obtained in the millisecond time as long as the characteristic input is given. With the optimization of algorithm and the improvement of computer hardware performance, the training time of neural network can be further shortened. Therefore, the calculation speed of the proposed method is significantly improved compared with the traditional method.

V. CONCLUSION

In this paper, the neural network is trained by selecting the appropriate electrical quantity as the characteristic quantity and combining with the corresponding load shedding strategy, so as to obtain the network model reflecting the mapping relationship between the system operation state and the load shedding strategy. Compared with the existing methods, it improves the calculation efficiency on the basis of ensuring the accuracy and has certain practical value. Training neural networks through unsupervised learning will be the next step of research.

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